JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY

COMPUTER NETWORKS (24B11CS311)

BTECH 5TH SEM

NETWORK LAYER

Q1. An IP datagram has arrived with the following information in the header (in hexadecimal):

45 00 00 54 00 03 00 00 20 06 00 00 7C 4E 03 02 B4 0E 0F 02

a. Are there anyoptions?
b. Is the packetfragmented?
c. What is the size of thedata?
d. Is a checksumused?
e. How many more routers can the packet travelto?
f. What is the identification number of thepacket?
g. What is the type ofservice?

Q2. a. Calculate the HLEN value if the total length is 1200 bytes, 1176 of which is data from the upper layer.
b. In a datagram, the M bit is zero, the value of HLEN is 5, the value of total length is 200, and the offset value is 200. What is the number of the first byte and number of the last byte in this datagram? Is this the last fragment, the first fragment, or a middle fragment?

Q3. a. Find the class of the following IP addresses:
208.34.54.12, 238.34.2.1, 242.34.2.8, 129.14.6.8
b. Find the netid and the hostid of the following IP addresses:
114.34.2.8, 132.56.8.6, 208.34.54.12, 251.34.98.5

Q4. In classless addressing, can two blocks have the same prefix length? Explain.

Q5. a. Given a fragmented datagram with an offset of 120, how can you determine the first and last byte number?
b. An IP datagram arrives with fragmentation offset of 0 and an *M* bit (more fragment bit) of 0. Is this a first fragment, middle fragment, or last fragment?
c. An IP fragment has arrived with an offset value of 100. How many bytes of data were originally sent by the source before the data in this fragment?

Q6. Consider sending a 2400-byte datagram into a link that has an MTU of 700 bytes. Suppose the original datagram is stamped with the identification number 422. How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation?

Q7. An organization is granted the block 16.0.0.0/8. The administrator wants to create 500 fixed-length subnets.

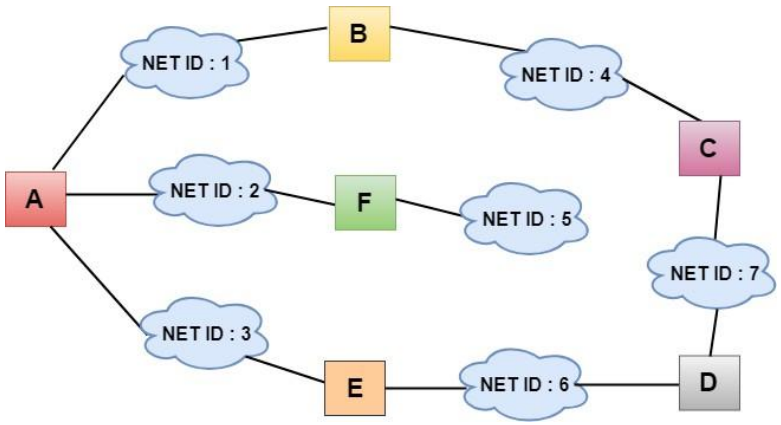
a. Find the subnetmask.
b. Find the number of addresses in each subnet.
c. Find the first and the last address in the firstsubnet.
d. Find the first and the last address in the last subnet (subnet500).

Q8. An ISP is granted a block of addresses starting with 150.80.0.0/16. The ISP wants to distribute these blocks to 2600 customers as follows:

a. The first group has 200 medium-size businesses; each needs approximately 128 addresses.
b. The second group has 400 small businesses; each needs approximately 16addresses.
c. The third group has 2000 households; each needs 4addresses.
d. Design the sub blocks and give the slash notation for each sub block. Find out how many addresses are still available after theseallocations.

Q9. An organization has a class C network: 200.1.1.0, and it wants to form subnets for 4 departments with the number of hosts as follows:

Subnet A: 72 hosts
Subnet B: 35 hosts
Subnet C: 20hosts
Subnet D: 18hosts
There are 145 hosts in total.

a. Provideapossiblearrangementof thenetworkaddressspace,togetherwiththerespective range of IP addresses for each subnet. Explain yourwork.
b. Suggest whattheorganizationmightdoifitneedstocreatethe5th subnet(SubnetE)with20 newhosts.

Q10. In the below figure, each cloud represents the network, and the number inside the cloud represents the network ID. All the LANs are connected by routers, and they are represented in boxes labeled as A, B, C, D, E, F. Distance vector routing algorithm simplifies the routing process by assuming the cost of every link is one unit. Therefore, the efficiency of transmission can be measured by the number of links to reach the destination. Build the Final routing tables of all the routers and show all the intermediate steps.